

Basic Logic Gates

INTENDED LEARNING OUTCOMES (ILOS):

After completing this chapter, the students will be able to:

- familiarize with the different logic gates
- list and explain the basic digital logic gates.
- use and explain the function
- ality of logic gates AND, NAND, OR, NOR, XOR, XOR, NOT)
- use the formal symbols for logic.
- construct truth table for propositional logic.

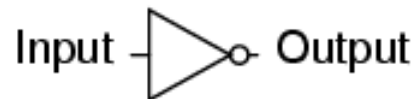
TRUTH TABLE

- is a standard way of representing the input/output relationships of a gate circuit, listing all the possible input logic level combinations with their respective output logic levels.

The Not or Inverter Gate

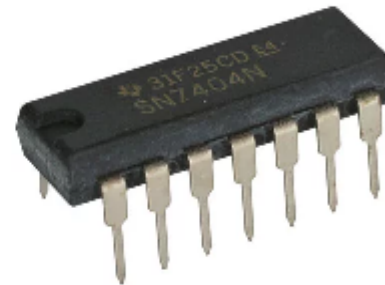
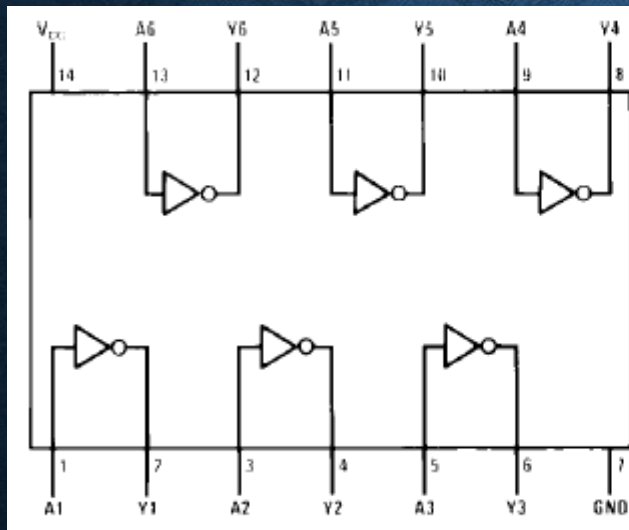
It is an electronic circuit that produces an *inverted version* of the input at its output. It is also known as an *Inverter*. If the input variable is A , the inverted output is NOT A . This is also shown as A' or $\bar{A}$ with a bar over the top.

NOT gate truth table



Input	Output
0	1
1	0

NOT GATE



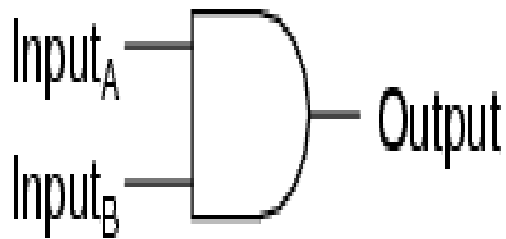
http://www.jameco.com/z/7404-Major-Brands-IC-7404-Hex-Inverter-DIP-14_49040.html

Connection diagram, and 74LS04 IC

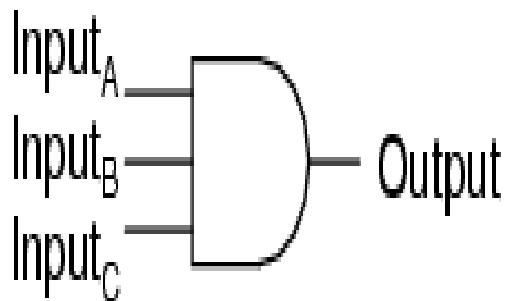
The AND gate

One of the easiest multiple-input gates to understand is the AND gate, so-called because the output of this gate will be "high" (1) if and only if *all* inputs (first input *and* the second input *and* . . .) are "high" (1). If any input(s) are "low" (0), the output is guaranteed to be in a "low" state as well.

2-input AND gate



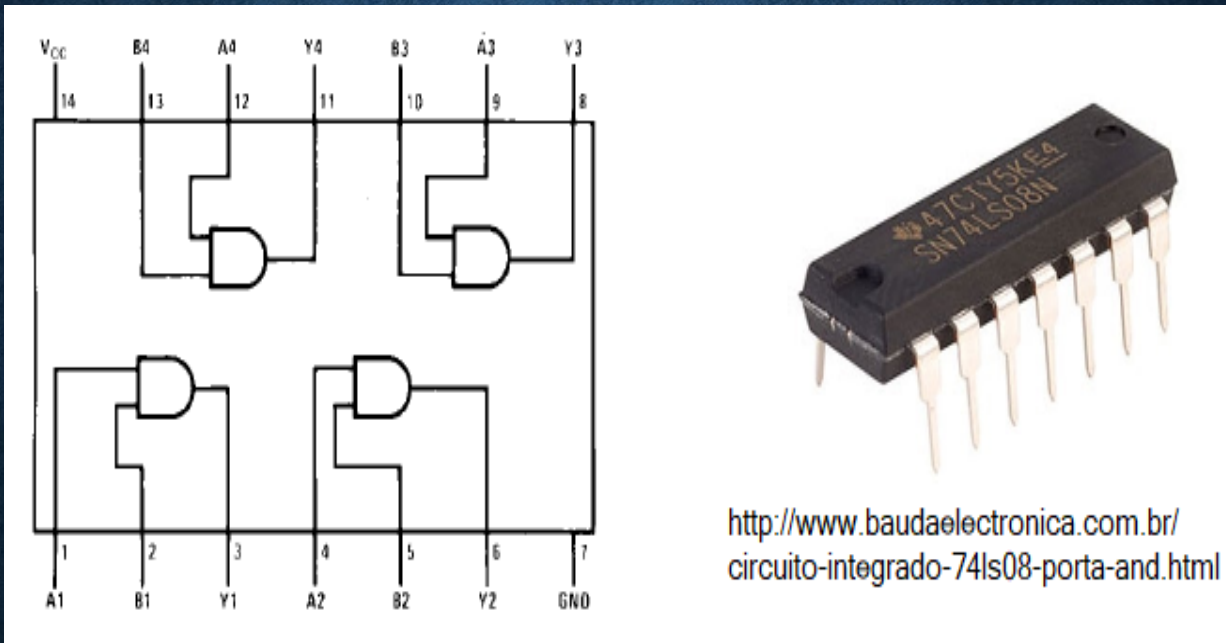
3-input AND gate



A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

AND gate symbol, Truth table,

AND GATE

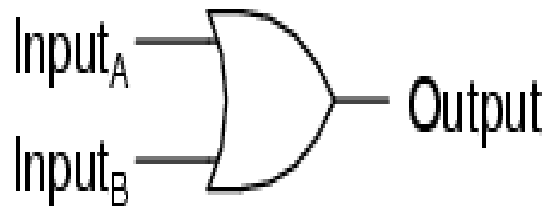


Connection diagram, and 74LS08 IC

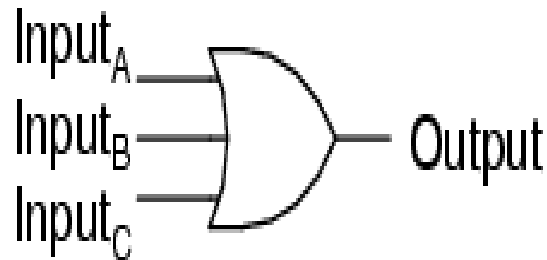
The OR gate

Our next gate to investigate is the OR gate, so-called because the output of this gate will be "high" (1) if *any* of the inputs (first input *or* the second input *or* . . .) are "high" (1). The output of an OR gate goes "low" (0) if and only if all inputs are "low" (0).

2-input OR gate

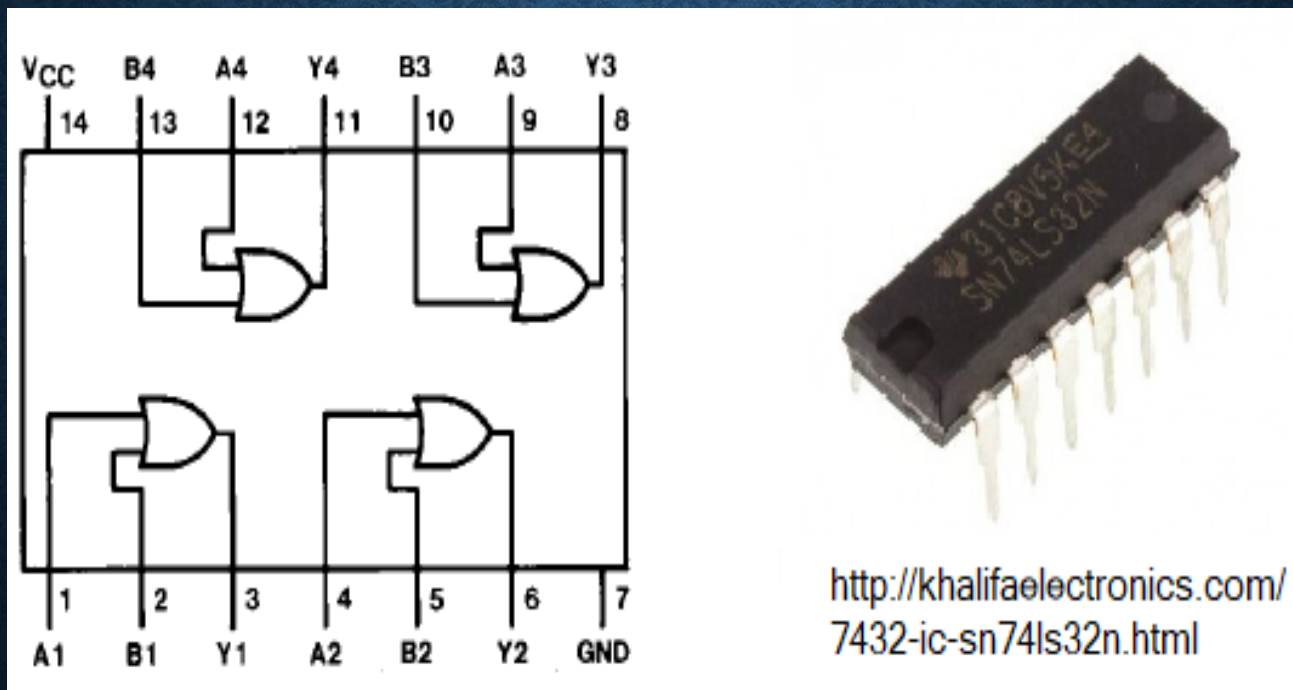


3-input OR gate



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

OR GATE

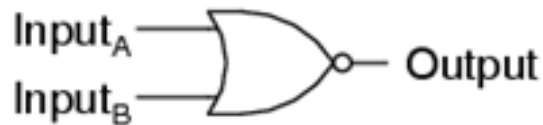


Connection diagram, and 74LS32 IC

The NOR gate

The NOR gate is an OR gate with its output inverted, just like a NAND gate is an AND gate with an inverted output.

2-input NOR gate

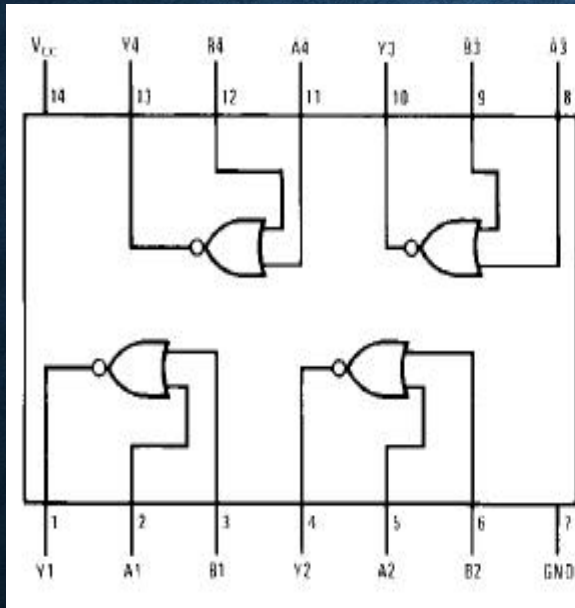


Equivalent gate circuit



A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

NOR GATE



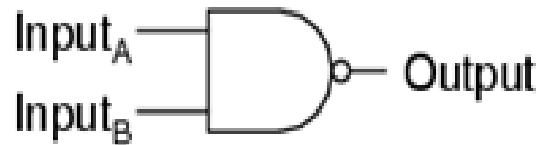
<http://www.electronicpro.com.pk/product/nor-gate-7401-ic/>

Connection diagram, and 74LS02 IC

The NAND gate

A variation on the idea of the AND gate is called the NAND gate. NAND gate behaves the same as an AND gate with a NOT (inverter) gate connected to the output terminal.

2-input NAND gate

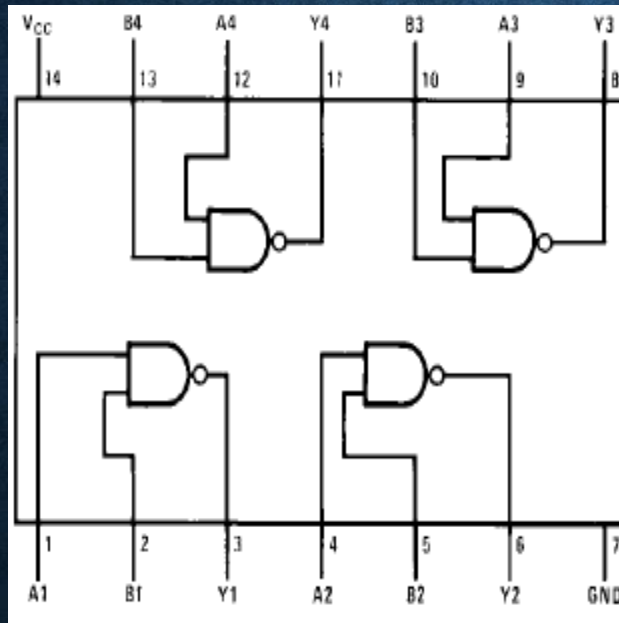


Equivalent gate circuit



A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

NAND GATE

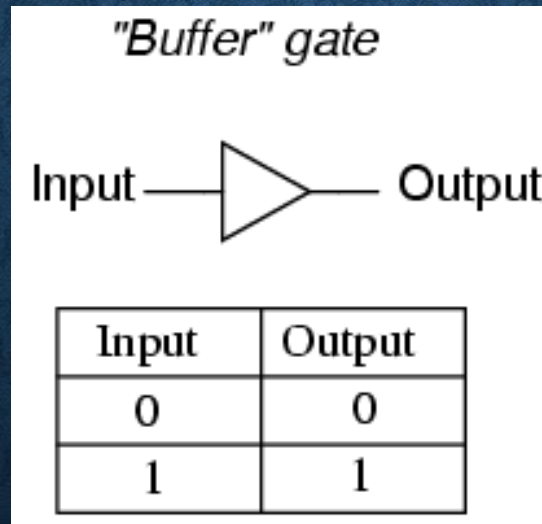


<https://www.kanbkam.com/sa/en/7400-ic-sn-74ls00n-quad-2input-nand-logic-gate-33005395>

Connection diagram, and 74LS00 IC

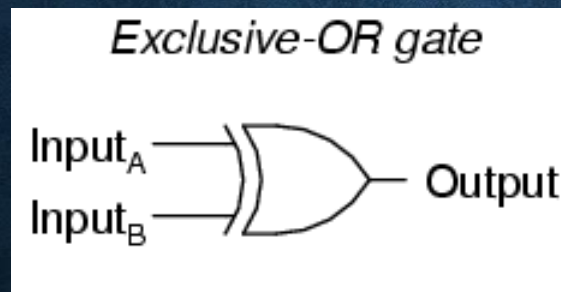
The "buffer" gate

If we were to connect two inverter gates together so that the output of one fed into the input of another, the two inversion functions would "cancel" each other out so that there would be no inversion from input to final output:



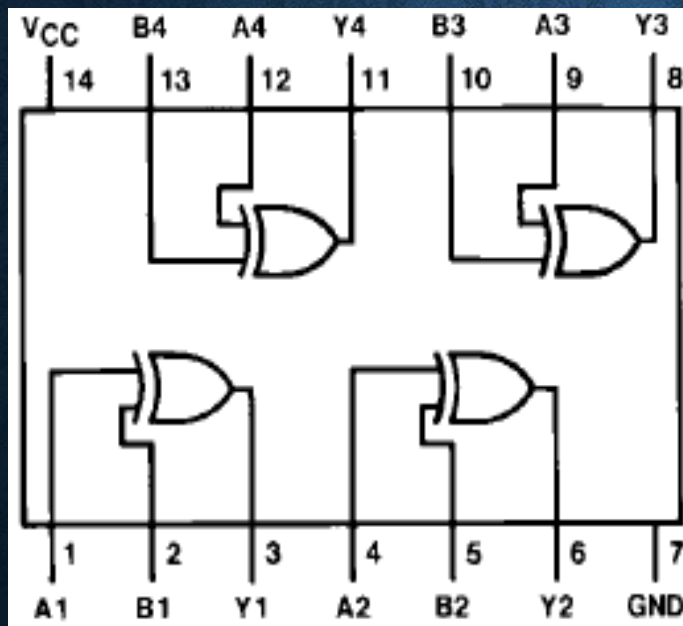
The Exclusive-OR gate

The Exclusive-OR gate, however, is something quite different. Exclusive-OR gates output a "high" (1) logic level if the inputs are at *different* logic levels, either 0 and 1 or 1 and 0. Conversely, they output a "low" (0) logic level if the inputs are at the *same* logic levels. The Exclusive-OR (sometimes called XOR) gate has both a symbol and a truth table pattern that is unique:



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

EXCLUSIVE OR GATE - XOR

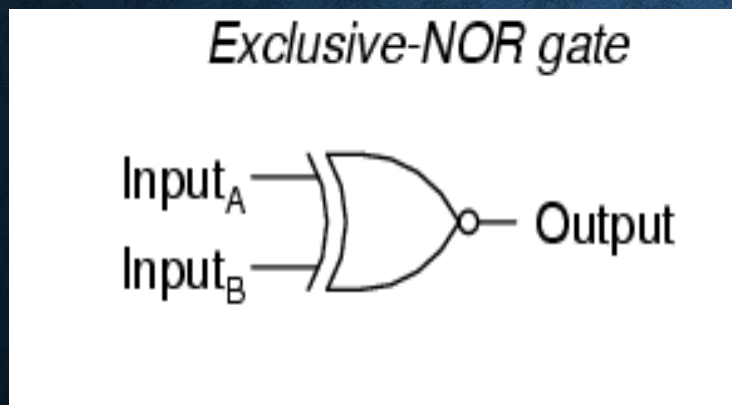


<http://www.electronicpro.com.pk/product-tag/7486-ic-in-pakistan>

Connection diagram, and 74LS86 IC

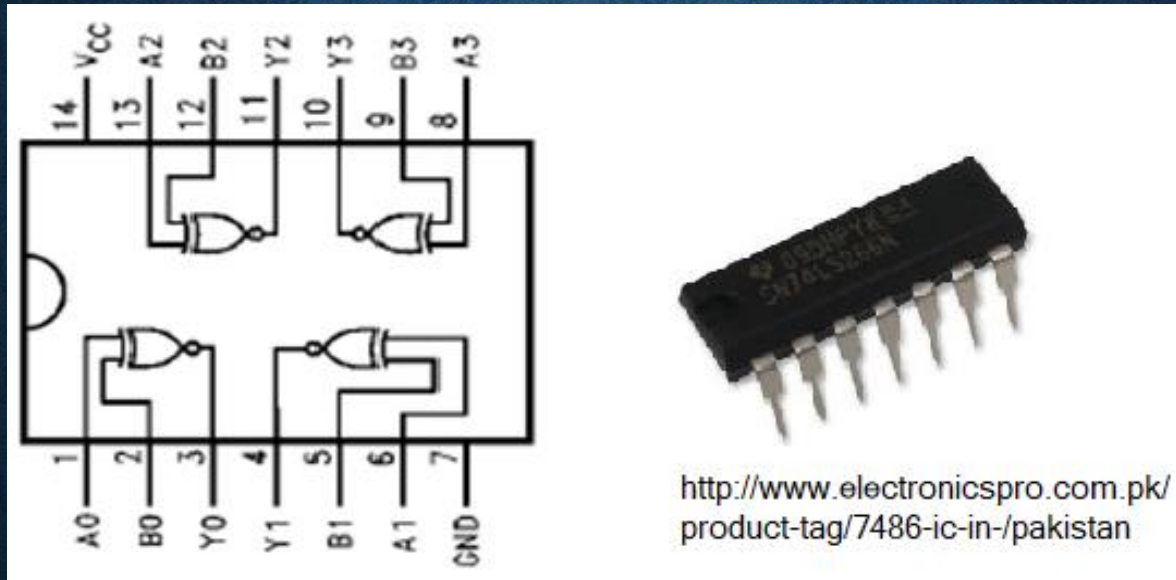
The Exclusive-NOR gate

Exclusive-NOR gate, otherwise known as the XNOR gate. It is equivalent to an Exclusive-OR gate with an inverted output. The truth table for this gate is exactly opposite as for the Exclusive-OR gate:



A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

EXCLUSIVE NOR GATE - XNOR



Connection diagram, and 74LS266 IC

THANK YOU FOR
LISTENING! 😊